

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

ANSWERS

1. Star Topology vs. Bus Topology

Cost

- Star topology is more expensive — needs a central switch/hub plus individual cable runs to every device
- Bus topology is cheaper — just one shared backbone cable, minimal hardware

Reliability

- Star: one node/cable failing doesn't affect others → high reliability
- Bus: a break anywhere in the backbone cable brings down the whole network → low reliability

Performance

- Star: central switch manages traffic, reduces collisions → consistent performance even as devices increase
- Bus: all devices share one medium → performance drops as more devices are added (more collisions, more congestion)

Real-world example

- Star topology → typical modern office LAN with a central switch
- Bus topology → old coaxial Ethernet setups (10BASE2), rarely used today

Conclusion

- Star topology costs more upfront but wins decisively on reliability and performance, which is why it's the industry standard today; bus topology survives only in small, low-budget, non-critical setups.

2. Mesh Topology vs. Star Topology

Fault Tolerance

- Mesh: every node connects to multiple others → multiple alternate paths if a link fails → very high fault tolerance
- Star: individual node failure is fine, but if the central hub/switch fails → entire network goes down → moderate fault tolerance

Redundancy

- Mesh: built-in redundancy at the link level
- Star: single point of failure at the center

Real-world example

- Mesh → military communication networks, core internet backbone routers
- Star → standard office/home networks
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Conclusion

- Mesh topology is the superior choice for mission-critical systems where downtime is unacceptable; star topology is a practical, cost-effective trade-off for everyday use where full redundancy isn't necessary.

3. Bus Topology vs. Mesh Topology

Installation Simplicity

- Bus: one backbone cable, minimal connections → quick and cheap to set up
- Mesh: every node connects to every other node → complex wiring, many ports/cables needed

Scalability of Complexity

- Mesh connections grow rapidly with more nodes — formula: $n(n-1)/2$ links
 - e.g., 10 devices in full mesh = 45 individual connections
- Bus stays simple regardless of small increases in devices.

Cost & Time

- Bus: low cost, fast installation
- Mesh: high cost, time-consuming, especially at scale

Real-world example

- Bus → small classroom/lab network
- Mesh → ISP backbone routers needing guaranteed redundancy

Conclusion

- Bus topology is ideal for small, simple, non-critical networks; mesh topology is reserved for environments where the redundancy is worth the installation complexity and cost.

4. Hybrid Topology vs. Star Topology

Scalability

- Hybrid: combines multiple topologies (e.g., star + bus, star + mesh) → can expand flexibly without disrupting the existing structure
- Star: scalable only up to the port capacity of the central device; beyond that, needs extra switches (can create bottlenecks)

Flexibility

- Hybrid: adapts to different organizational needs across departments/locations
- Star: simple and uniform, but rigid once it outgrows the central device

Real-world example

- Hybrid → large enterprise/campus networks — star topology inside each building, connected via a backbone (bus/mesh) between buildings
- Star → single-office or small business network

Conclusion

- Hybrid topology suits large, growing organizations needing flexibility across sites; star topology is better for smaller, single-location networks where simplicity outweighs the need for scale.

5. Maintenance Comparison — Star vs. Mesh vs. Bus vs. Hybrid

- **Star** — Easiest to maintain
 - Faults isolated to a single node/cable
 - Quick to diagnose (swap cable/port)
- **Bus** — Harder to maintain
 - Backbone cable faults affect entire network
 - Difficult to physically locate the break
- **Mesh** — Most difficult to maintain
 - Dense web of connections makes fault-tracing time-consuming
 - Requires skilled troubleshooting despite high reliability
- **Hybrid** — Moderate maintenance complexity
 - Flexible, but requires monitoring multiple combined topologies
 - Needs more experienced network administrators

Conclusion

- Maintenance difficulty generally increases with structural complexity: **Star (easiest)** → **Bus/Hybrid (moderate)** → **Mesh (hardest)**. Network designers must balance reliability benefits against the maintenance effort each topology demands.